

Zachary Boice

Curriculum Vitae

Geneseo, NY — zboice05@gmail.com — Personal Website

Education & Academic Employment

B.S., Physics, B.A., Mathematics

Expected: May 2027

SUNY Geneseo

Minor in college honors program, microcredential in Computational and Theoretical physics, senior capstone on gravitational waves advised by Dr. Thomas Osburn. Skills developed: advanced knowledge of python (including numpy, scipy, robust visualizations using matplotlib, and understanding of object-oriented programming), advanced knowledge of Mathematica (specifically pertaining to numerically solving ODEs and PDEs), proficiency in L^AT_EX, proficiency in Excel, working knowledge of HTML and CSS, basic knowledge of Matlab and JavaScript.

Research Assistant

Time Period: June 2024 - Present

SUNY Geneseo

Physics

Undergraduate researcher working under the supervision of Dr. Thomas Osburn. Developed code as part of work to calculate the gravitational self-force and emitted waves via a puncture and effective source method in the frequency-domain. Working to create a framework to efficiently solve the Regge-Wheeler equation in the time-domain with highly accurate boundary conditions.

Student Assistant

Time Period: August 2023 - Present

SUNY Geneseo

Chemistry Stockroom

Coordinated the operation of Bio-, Analytical, Physical, and Introductory chemistry labs by preparing necessary chemicals. Responsible for delegating tasks to ensure unstable chemicals are prepared safely and in a timely manner. Experienced in hazardous waste management and preparation of chemicals at all ranges of concentration and volume.

Papers

Reviewed Articles

Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc.

Class Projects

- Intermediate Laboratory - Analysis of Chladni Plate Patterns, **Z. Boice**, C. Robb (Spring 2026)
- Optics - Hamiltonian Optics and higher-level structures, **Z. Boice** (Spring 2026)
- Linear Programming and Operations Research - Intensity Modulated Radiation Therapy, **Z. Boice**, M. Moroz (Spring 2026)
- Numerical Analysis I - The Galerkin Method, **Z. Boice**, C. Robb (Fall 2025)

GRANTS

External Grants

Funded

Proposal Title:

Funding Agency:

Role Status:

Percentage of Attributed Effort:

Total direct costs and annual budget:

Coverage Period:

Pending

Proposal Title:
Funding Agency:
Role Status:
Percentage of Attributed Effort:
Total direct costs and annual budget:

Not Funded (Optional but recommended)

Proposal Title:
Funding Agency:
Role Status:
Percentage of Attributed Effort:

Internal Grants

Geneseo Summer Fellowship

Proposal Title: Developing a novel method to calculate gravitational waves emitted by highly eccentric Black Hole binaries
Funding Agency: SUNY Geneseo Foundation
Budget: \$3000 (with an additional \$600 for Geneseo's Rhodes award, given annually to the summer fellowship applicant whose proposal is the highest ranked among submitted projects)
Coverage Period: Summer 2026

We seek to calculate the energy flux emitted by an eccentric extreme mass-ratio inspiral. To do this, we solve the Regge-Wheeler equation using the method of lines and numerically integrating; a fourth-order finite difference method is used to increase accuracy. We represent the source by enforcing a jump condition across an internal boundary. We also make use of a hyperboloidal slicing and compactification scheme, based on the work of Thornburg and Wardell (2017). We seek to create a coordinate transformation that shifts the space around an effectively stationary source, which will greatly simplify calculations. With a fourth-order finite difference method and exact compactified boundary conditions, we hope to achieve results among the most accurate time domain calculations.

Frank Vafier '74 Ambassadorship in Leadership

Proposal Title: Addressing Light Pollution at SUNY Geneseo
Funding Agency: SUNY Geneseo
Budget: \$5000 (awarded jointly to ZLB, Gavin George, and Christopher Desiderio)
Coverage Period: Calendar year 2026

The level of light pollution in and around SUNY Geneseo's campus poses a threat to public health and safety and to the habitats of native species, as well as obscures the natural beauty of the night sky for campus and village community members. We seek to reduce light pollution by targeting factors like overlighting and the use of cool-color lights. We will design an inexpensive covering that can be retrofitted on existing light fixtures, minimizing the amount of light that is directed into the sky instead of illuminating walkways.

Presentations

Talks

- Developing a Novel Method to Calculate Gravitational Waves Emitted by Highly Eccentric Black Hole Binaries, **Z. Boice** - Geneseo GREAT Day 2026
- Addressing Light Pollution at SUNY Geneseo, C. Desiderio, **Z. Boice**, G. George- Geneseo GREAT Day 2026
- Numerical Solutions to PDEs (with Astrophysical Applications), **Z. Boice** - Geneseo GREAT Day 2025

Posters

- Accurate Gravitational Wave Calculations from the Regge-Wheeler Equation in the Time Domain, **Z. Boice** - Presented at the APS Global Physics Summit March 2025 and GREAT Day 2025
- Towards Extreme Mass-Ratio Inspiral Calculations in the Time Domain Using Hyperboloidal Slicing and Compactification, J. McNamara, **Z. Boice**, T. Osburn - Presented at the APS Global Physics Summit March 2025 and GREAT Day 2025

Teaching Assistance

Courses

- Calculus II Supplemental Instructor (Spring 2025) - Helped Geneseo students taking Calculus II by running problem sessions and facilitating discussions outside of class time. Created various graphics and demos to improve understanding.

Tutoring

- Physics Department (Fall 2025 - Present) - Available two hours weekly to support younger students as they go through the introductory sequence of classes.
- Private Tutoring (Fall 2023 - Fall 2024) - Worked with peers one-on-one to explain concepts in Calculus I and II. Sought out by parents to raise student's grades above failing, 100% success rate (four students total, none failed).

Service

Service to the Department

- Physics and Astronomy Club, *President (Fall 2024 - Present)* - Coordinated biweekly events related to Physics and Astronomy, generally targeted at a popular audience (see below). Additional events include an annual picnic, semesterly department social event (pool and pizza), bringing in a counselor to discuss mental health and wellness, providing guidance and advising help for first-year students, and a physics-based game show.

Outreach

- Planetarium Shows - Developed content for several shows using the SUNY Geneseo planetarium (attendance of roughly 30 students per event). Included researching various cultures to teach about non-western views of the sky, facilitating an interdisciplinary writing space-centric writing contest, and generally coordinating logistics.
- Stargazing Nights - Coordinated use of the Physics and Astronomy Club's telescope to view the night sky, with some education about the objects we view (roughly 20 students per event).
- Geneseo Recruitment - Hosted prospective Physics majors for a day-long visit to campus. Later, selected as an ambassador to be interviewed for short form content advertising SUNY Geneseo and its Physics and Astronomy programs.

Awards and Scholarships

Departmental

- Physics - Excellence in Physics (annual scholarship, with additional awards Spring 2025 and Spring 2026), Service to the Department (Spring 2026)
- Mathematics - Excellence in Mathematics (separate scholarships awarded in Spring 2025, and Spring 2026), Second highest Putnam score in the department (Spring 2026)